

22 Dec 2025

PS Seminar 8

Function indicator

Funcția indicator

$$1_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

$$f: A \rightarrow B$$

$$\text{supp } f = \{x \in A \mid f(x) \neq 0\}$$

$$c = \frac{3}{8}$$

deoarece pt aceasta const
f este funcție de densitate de prob.

Obs Dacă f este densitate de prob.

$$\text{supp } f = \{x \in \mathbb{R} \mid f(x) > 0\}$$

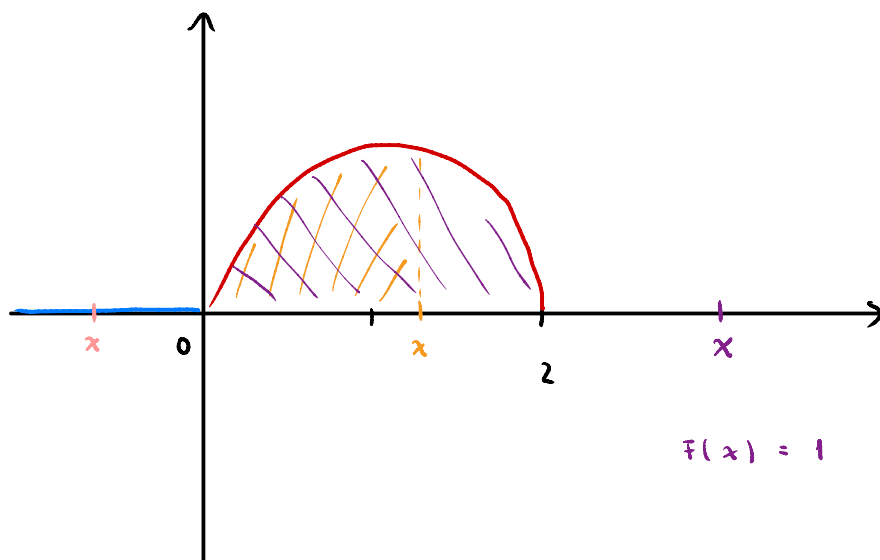
$$\cong f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} c(4x - 2x^2), & x \in (0, 2) \\ 0, & \text{altfel} \end{cases}$$

"
supp f

$$= c \cdot (4x - 2x^2) \cdot 1 \Big|_{(0,2)}^{(x)}$$

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad (\Rightarrow) \quad \int_{\text{supp } f} f(x) dx = 1$$

$$E(x) = \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_{\text{supp } f} x \cdot f(x) dx$$



$$F(x) = 1$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

! Din culegere ASE capitolul 2.2